Akash Dasarraju

(469) 428-0652 | dasarrajuakash@gmail.com | sky1219.github.io
linkedin.com/in/akash-dasarraju-143520240 | github.com/Sky1219

ML Systems & LLM Inference Optimization | Seeking Summer 2027 internship

Personal Projects (Solo)

hipinfer | *C++, HIP/CUDA* | github.com/Sky1219/hipinfer | sky1219.github.io/hipinfer | Colab 2026 – Present

- Reached 92–95% of cuBLAS speed on Turing with a from-scratch GEMM, in an LLM inference engine I built in HIP that runs on NVIDIA and AMD GPUs; matches Hugging Face output within 0.003%
- Fastest single-request FP16 decode measured on all 5 Llama/Qwen models tested on an RTX 2080 Ti: 6–17% faster than llama.cpp on 1–3B models, at 81–87% of its memory-bandwidth limit; also beat vLLM on an RTX A6000
- Added multi-GPU tensor parallelism with a custom all-reduce 1.5x faster than NCCL on small messages: 1.55–1.75x faster decode on 2 RTX 2080 Tis, and runs 7–8B models (Llama-3.1-8B, Mistral-7B) too large for one card

SubCache | *Python, PyTorch* | github.com/Sky1219/SubCache | sky1219.github.io/SubCache 2026

- Built an open-source KV-cache compression benchmark: on long coding-agent sessions, keeping 25% of the cache fit 3x more requests on one GPU, raising throughput 2.8x (64% lower cost/token) for a 5% perplexity increase
- Benchmarked 9 eviction policies, including published baselines H2O and R-KV, on real coding-agent sessions, then on LongBench, needle-in-a-haystack retrieval, and MATH-500 with Llama-3.1-8B
- Designed a submodular eviction method that beat H2O by 1.1–2.5% on repetitive agent sessions, then profiled it and pinpointed the bottleneck: choosing what to evict took 71% of decode time (under 1% for H2O)

mnk-bench | *Python, PyTorch* | github.com/Sky1219/mnk-bench | sky1219.github.io/mnk-bench 2026

- Built an open-source benchmark of 10 game-playing AI methods, from classic search to self-play networks and LLMs (118 tests, CI); my 39k-parameter AlphaZero-style network never lost to a perfect tic-tac-toe player

Research Experience

Researcher, CARAML Lab (Prof. Rishabh Iyer) Mar. 2026 – Present

Submodular ML lab; I build and run the experiments for two in-prep papers, guided by PhD students *Richardson, TX*

- Developing submodular training losses that use hard negatives to teach CLIP-style vision-language models object–attribute relationships; built and evaluated the losses and generated training data (targeting CVPR)
- Developing a submodular method that picks which past conversation turns an LLM agent keeps under a token budget, compared against top-k and MMR, self-hosting Qwen2.5-32B with vLLM on an A6000 (targeting ICML)

Experience

AI Tools Developer, SoyMedical *(online medical supplies distributor)* Summer 2026

- Built an AI agent tool on OpenAI's Codex CLI that researches a product, compares Amazon prices, writes marketing copy, and returns a structured JSON order recommendation with a plain-language explanation
- Now a standard step in the company's ordering workflow: made product evaluation 2x faster across hundreds of products; an experienced buyer reviews every recommendation and agrees with about 70%

Leadership

HackUTD, Organizer (Leadership Team) Jan. 2026 – Present

The University of Texas at Dallas *Richardson, TX*

- Organize a 1,300-participant hackathon; built a 3D campus map (three.js) with 3D modeling and animation
- Tripled workshop attendance (30 to 100+) by redesigning topics for 10 AI workshops (led 5, assisted 5)

Technical Skills

Languages & Tools: Python, C++, Java; CUDA, HIP/ROCm, NCCL, PyTorch, vLLM, Hugging Face
GPUs: NVIDIA Turing & Ampere, AMD

Education

The University of Texas at Dallas Richardson, TX

B.S. in Computer Science, CS Squared Honors Program; GPA: 3.84 *Aug. 2025 – May 2029 (expected)*

- Coursework: Algorithms & Data Structures (Hon.), Statistics for ML (Hon.), Linear Algebra, Comp. Architecture
- National Merit Finalist; ACT 36